

Olivocochlear projections contribute to superior intensity coding in cochlear nucleus small cells

Adam Hockley, Calvin Wu, and Susan Shore

DOI: 10.1113/JP282262

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The referees have opted to remain anonymous.

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(Note: With the exception of the correction of typographical or spelling errors that could be a source of ambiguity, letters and reports are not edited. Depending on transfer agreements, referee reports obtained elsewhere may or may not be included in this compilation. Referee reports are anonymous unless the Referee chooses to sign their reports.)

Dear Professor Shore,

Re: JP-RP-2021-282262 "Cochlear nucleus small cells use olivocochlear collaterals to encode sounds in noise" by Adam Hockley, Calvin Wu, and Susan Shore

Thank you for submitting your manuscript to The Journal of Physiology. It has been assessed by a Reviewing Editor and by 2 expert Referees and I am pleased to tell you that it is considered to be acceptable for publication following satisfactory revision.

Please advise your co-authors of this decision as soon as possible.

The reports are copied at the end of this email. Please address all of the points and incorporate all requested revisions, or explain in your Response to Referees why a change has not been made.

NEW POLICY: In order to improve the transparency of its peer review process The Journal of Physiology publishes online as supporting information the peer review history of all articles accepted for publication. Readers will have access to decision letters, including all Editors' comments and referee reports, for each version of the manuscript and any author responses to peer review comments. Referees can decide whether or not they wish to be named on the peer review history document.

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I hope you will find the comments helpful and have no difficulty returning your revisions within 4 weeks.

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I look forward to receiving your revised submission.

If you have any queries please reply to this email and staff will be happy to assist.

Yours sincerely,

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EDITOR COMMENTS

Reviewing Editor:

This is a very well executed manuscript reporting novel and interesting finding. The quality of the data and analysis is very good and the manuscript is well written. Both reviewers are very positive but have also raise several points that have to be addressed.

While revising the manuscript, please make sure you also address the following points:

- 1) Line 44, Abstract: define SCC
- 2) Line 54-61, Key Points. It would be useful to have 1-2 points in the middle explaining the key results more widely accessible. The third point reaches a conclusion without having explained how. You still have plenty of space in this section.
- 3) The method of euthanasia and the end point of the procedures should be clearly stated.
- 4) The Statistical section (Methods) should state that all average data are reported as mean $\pm$ SD (no SEM).
- 5) The authors should clearly specify the "n" (number of observations and/or biological replica) when quoting a P value and/or mean and SD (unless clearly indicated in the Figure legend).
- 6) Individual data points should be included in all graphs showing averages (e.g. Figure 3, 4, 5, 7)
- 7) Average data reported in the text and Figures should indicate mean $\pm$ SD (instead of SEM).
- 8) The Statistical Summary should be provided
- 9) Supplementary Figures (and Table) are not allowed, unless when used to report individual data points on graphs with $n > 30$. As such, you can either include them in the main manuscript or please remove them.

Senior Editor:

Comments for Authors to ensure the paper complies with the Statistics Policy:
Please see RE comments

REFeree COMMENTS

Referee #1:

The manuscript "Cochlear nucleus small cells use olivocochlear collaterals to encode sounds in noise." by Hockley et al. investigates sound encoding properties of small cells of the cochlear nucleus. This study is overall well written and used efficient recording and analysis methods to generate large sample sizes, building confidence in the findings. The authors show that among these findings, muscarinic receptor activity contributes to high intensity stimulus responses; this effect plays a role in the accurate representation of sound intensity among these neurons. This representation is robust in the presence of noise. I think this paper will be of interest to a broad audience within auditory neuroscience, and the cholinergic findings are very interesting. Here I present both major and minor critiques presented here in order to improve the manuscript.

1) The title of this paper would suggest that the MOC influence on intensity coding in the presence of noise was actually tested. It was not. This statement can only be asserted based on the finding that RLFs in the absence of noise have a muscarinic component- and that the population responses are not appreciably different in noise- therefore this assertion is not based on a properly controlled experiment. There are lots of interesting findings in this paper to support a compelling revision of the title, but the current title is not acceptable.

2) Line 77 "In humans, the SCC occupies a larger proportion of the CN than in rodents, and is therefore poised to play a major role in central mechanisms of speech perception (Moore & Osen, 1979)." This statement far exceeds credibility with respect to evidence, a more conservative and explicitly speculative approach would be more palatable.

3) Fig 1A, recommend color/symbol choices that better distinguish small cells from onset population as depicted

4) Line 287 Reference to Goyer et al. 2016 is notably missing, as this study illustrates findings that are highly similar to the current study, albeit in another CN population.

5) "These findings demonstrate that the primary action of cholinergic input on small cells and choppers is to increase sound-driven responses above 60 dB, resulting in a widened dynamic range that is essential for intensity coding." --The above comment is too broad, it only applies to muscarinic influence. Unless nicotinic antagonists are also applied one cannot so definitively use the comprehensive "cholinergic" category

6) Fig 2 and associated text. Single sample shows no borders of the VNTB and there seems to be broader involvement of the MNTB. A higher standard could be easily reached with a counterstain so that nuclear borders can be convincingly demarcated with respect to the injection site.

7) A more thorough description of the machine learning methodology should be included in the methods section or the code should be available as a supplement in sufficient detail to allow for objective evaluation.

8) Line 253 These statements must be referenced

9) Overall the study benefits from large sample sizes which provide statistically robust population data. However, the whole presentation would be more convincing if the authors would show examples of individual unit behavior in several places, especially with regard to pharmacological or electrical stimulation experiments. In the case of the former, a major concern is that there are no examples of recovery from treatment in any experiment despite seemingly ample opportunity built into the efficient experimental design.

10) The VNTB stimulation experiments are compelling, but would be much more conclusive if they were also conducted in the presence of atropine at the recording site. The current results rely on an assumption that the only neurons that were activated by stimulation are the cholinergic MOCs.

Referee #2:

In the manuscript Cochlear nucleus small cells use olivocochlear collaterals to encode sounds in noise (JP-RP-2021-282262) by Hockley et al, the authors use in vivo neuronal recordings and sound stimulation to probe the activation and function of neurons of the cochlear nucleus small cell cap (SCC). The activation of these neurons in response to sound and their potential role in auditory perception has been incompletely characterized, and the authors hypothesize that these neurons possess response characteristics that make them particularly well-suited to processing speech, in part due to influence from MOC neurons that excite SCC neurons at high sound levels when the cochlear afferent input is suppressed (by MOC activity in the periphery).

The authors used in vivo multielectrode recordings of SCC neurons, electrical stimulation of MOC efferent neurons, fluorescent neuron tract tracing, sound exposure, pharmacology, and auditory function testing to demonstrate that SCC neurons have bi-directional circuits with neurons of the VNTB where MOC neurons reside, improved intensity encoding at higher sound levels that is sensitive to both cholinergic pharmacology and electrical stimulation of the VNTB to activate MOC circuits, superior intensity coding in the presence of background sound, and more accurate responses to amplitude modulated tones at the lower sound frequencies used for encoding vocalizations. Overall the work is concisely written, the figures are clear, and the statistics are appropriate. No single piece of evidence proves that the SCC neurons provide direct synaptic inputs from the MOC neurons as the histology is specific for neurons in the VNTB but not necessarily for the MOC neurons within the VNTB, but the combined affect of cholinergic receptor modulation and electrical stimulation of the VNTB, where the MOC neurons are the only cholinergic cells, is convincing. The work also provides important insight into the function of the SCC neurons as potentially being involved in perception of speech.

Minor comments:

- Line 218, 'dorsoventrally', should this perhaps read 'dorsolaterally'?
- Line 240, "that SCC neurons form a circuit with MOC neurons" is too strong wording for this histological study, but it does show that SCC neurons form a circuit with neurons of the VNTB where the MOC somata are.
- Line 257 - capitalize "Small" at beginning of line
- Figure 4 - why is there no effect of atropine on PL cells in panel A, when there is a clear effect of VNTB stimulation in panel B? Please address.
- Line 347 - please elaborate on the methods for the application of spike trains - how applied and where?
- Line 373 - edit wording of the panel B legend, word incorrect or missing
- Figure 6 data - In panel A and D, the SCs show a number of responses classified as 80 dB, but with an actual dB anywhere from 0 to 80. Please address this, it seems to actually suggest poor intensity coding or the SCs in many instances. What are the criteria for "correct" encoding of intensity?
- Figure 7 results text beginning line 382 - is there an example calculation that was missing? The text refers to an example at 32 kHz, 12% depth, but this is just one point in the fourth panel. Also, the letters are missing from the figure.

- Missing citation of Romero and Trussell 2021, eLife, line 467 - descending input from IC to MOC

END OF COMMENTS

Confidential Review

11-Aug-2021

REVISION CHECKLIST:

- Article file, including any tables and figure legends, must be in an editable format (eg Word)
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EDITOR COMMENTS

Reviewing Editor:

This is a very well executed manuscript reporting novel and interesting finding. The quality of the data and analysis is very good and the manuscript is well written. Both reviewers are very positive but have also raise several points that

have to be addressed.

Thank you for the positive feedback, we have addressed all comments on this manuscript and conducted more experiments where necessary.

While revising the manuscript, please make sure you also address the following points:

1) Line 44, Abstract: define SCC

Added

2) Line 54-61, Key Points. It would be useful to have 1-2 points in the middle explaining the key results more widely accessible. The third point reaches a conclusion without having explained how. You still have plenty of space in this section.

Key points section has been expanded

3) The method of euthanasia and the end point of the procedures should be clearly stated.

Added

4) The Statistical section (Methods) should state that all average data are reported as mean $\pm$ SD (no SEM).

Added

5) The authors should clearly specify the "n" (number of observations and/or biological replica) when quoting a P value and/or mean and SD (unless clearly indicated in the Figure legend).

Corrected throughout

6) Individual data points should be included in all graphs showing averages (e.g. Figure 3, 4, 5, 7).

As $n > 30$, data points have not been plotted in the figure but the raw datasets for the RLFs in figs 3,4,5&7 have been uploaded as 'Supporting Information for online publication', named "CellTable_RLFs"

7) Average data reported in the text and Figures should indicate mean $\pm$ SD (instead of SEM).

This has been corrected, all figures and reporting of data is now SD

8) The Statistical Summary should be provided

This has been provided

9) Supplementary Figures (and Table) are not allowed, unless when used to report individual data points on graphs with $n > 30$. As such, you can either include them in the main manuscript or please remove them.

Supplemental figure & table have been removed from the manuscript.

REFeree COMMENTS

Referee #1:

The manuscript "Cochlear nucleus small cells use olivocochlear collaterals to encode sounds in noise." by Hockley et al. investigates sound encoding properties of small cells of the cochlear nucleus. This study is overall well written and used efficient recording and analysis methods to generate large sample sizes, building confidence in the findings. The authors show that among these findings, muscarinic receptor activity contributes to high intensity stimulus responses; this effect plays a role in the accurate representation of sound intensity among these neurons. This representation is robust in the presence of noise. I think this paper will be of interest to a broad audience within auditory neuroscience, and the cholinergic findings are very interesting. Here I present both major and minor critiques presented here in

order to improve the manuscript.

Thank you for the positive note and your comments, we have addressed all of the concerns. Experiments studying the recovery from atropine-infusion effects and combining VNTB stimulation have been added.

1) The title of this paper would suggest that the MOC influence on intensity coding in the presence of noise was actually tested. It was not. This statement can only be asserted based on the finding that RLFs in the absence of noise have a muscarinic component- and that the population responses are not appreciably different in noise- therefore this assertion is not based on a properly controlled experiment. There are lots of interesting findings in this paper to support a compelling revision of the title, but the current title is not acceptable.

Title has been changed from "Cochlear nucleus small cells use olivocochlear collaterals to encode sounds in noise" to "Olivocochlear projections contribute to superior intensity coding in cochlear nucleus small cells"

2) Line 77 "In humans, the SCC occupies a larger proportion of the CN than in rodents, and is therefore poised to play a major role in central mechanisms of speech perception (Moore & Osen, 1979)." This statement far exceeds credibility with respect to evidence, a more conservative and explicitly speculative approach would be more palatable.

This has been reworded as: "In humans, the SCC occupies a larger proportion of the CN than in rodents (Moore & Osen, 1979), suggesting that the SCC may have a greater role in central auditory processing in humans."

3) Fig 1A, recommend color/symbol choices that better distinguish small cells from onset population as depicted

Onset color has been changed from orange to yellow to give clearer separation

4) Line 287 Reference to Goyer et al. 2016 is notably missing, as this study

illustrates findings that are highly similar to the current study, albeit in another CN population.

A reference has been added to discussion "Spherical bushy cells have also been shown to receive respond to nicotinic and muscarinic cholinergic inputs, with muscarinic inputs providing long-term membrane resting potential modulation (Goyer et al., 2016)."

5) "These findings demonstrate that the primary action of cholinergic input on small cells and choppers is to increase sound-driven responses above 60 dB, resulting in a widened dynamic range that is essential for intensity coding." -- The above comment is too broad, it only applies to muscarinic influence. Unless nicotinic antagonists are also applied one cannot so definitively use the comprehensive "cholinergic" category.

Reworded to be specific for muscarinic input to small cells and choppers. "These findings demonstrate that the primary action of muscarinic cholinergic input on small cells and choppers is to increase sound-driven responses above 60 dB, resulting in a widened dynamic range that is essential for intensity coding."

6) Fig 2 and associated text. Single sample shows no borders of the VNTB and there seems to be broader involvement of the MNTB. A higher standard could be easily reached with a counterstain so that nuclear borders can be convincingly demarcated with respect to the injection site.

We agree with the reviewer that further tract tracing experiments would be helpful. However, we believe further experiments are beyond the scope of this figure, which aims only to confirms previous findings of Ye et al. (2000) and Ryan et al. (1990). We apologize that our original Figure 2 did not show VNTB border and have now added it to the figure. Some of this example injection did spread to the MNTB, but this would not be expected to produce the observed puncta staining in the CN, as no projections from MNTB to CN are known. Also, no projections from the SCC to MNTB would be expected. This has been commented on at line 242.

7) A more thorough description of the machine learning methodology should be included in the methods section or the code should be available as a supplement in sufficient detail to allow for objective evaluation.

The machine learning data and model have been added to supplemental data.

8) Line 253 These statements must be referenced

References added: "The high-fidelity intensity coding and absence of RLF plateaus in small cells is likely due to their combined input from low SR ANFs (Liberman, 1991) and excitatory input from MOCs at high sound levels neurons (Benson et al., 1996; Benson & Brown, 1990; Ryan et al., 1990)."

9) Overall the study benefits from large sample sizes which provide statistically robust population data. However, the whole presentation would be more convincing if the authors would show examples of individual unit behavior in several places, especially with regard to pharmacological or electrical stimulation experiments. In the case of the former, a major concern is that there are no examples of recovery from treatment in any experiment despite seemingly ample opportunity built into the efficient experimental design.

An additional experiment studying recovery from atropine infusion was conducted. Recordings were made up to 3h post-infusion. A single example of recovery from atropine in a small cell has been added as Fig 4B.

10) The VNTB stimulation experiments are compelling, but would be much more conclusive if they were also conducted in the presence of atropine at the recording site. The current results rely on an assumption that the only neurons that were activated by stimulation are the cholinergic MOCs.

An experiment combining VNTB stimulation and atropine infusion produced the single example of a small cell in Fig 4D. This shows that electrical stimulation increased activity of the unit, which was blocked by the subsequent infusion of atropine.

Referee #2:

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The authors used in vivo multielectrode recordings of SCC neurons, electrical stimulation of MOC efferent neurons, fluorescent neuron tract tracing, sound exposure, pharmacology, and auditory function testing to demonstrate that SCC neurons have bi-directional circuits with neurons of the VNTB where MOC neurons reside, improved intensity encoding at higher sound levels that is sensitive to both cholinergic pharmacology and electrical stimulation of the VNTB to activate MOC circuits, superior intensity coding in the presence of background sound, and more accurate responses to amplitude modulated tones at the lower sound frequencies used for encoding vocalizations. Overall the work is concisely written, the figures are clear, and the statistics are appropriate. No single piece of evidence proves that the SCC neurons provide direct synaptic inputs from the MOC neurons as the histology is specific for neurons in the VNTB but not necessarily for the MOC neurons within the VNTB, but the combined affect of cholinergic receptor modulation and electrical stimulation of the VNTB, where the MOC neurons are the only cholinergic cells, is convincing. The work also provides important insight into the function of the SCC neurons as potentially being involved in perception of speech.

Thank you for these comments, we have addressed all below.

Minor comments:

- Line 218, 'dorsoventrally', should this perhaps read 'dorsolaterally'?

To remove confusion, wording has been changed from "Small cells, located dorsoventrally between DCN and VCN..." to "Small cells, located between DCN and VCN..."

- Line 240, "that SCC neurons form a circuit with MOC neurons" is too strong wording for this histological study, but it does show that SCC neurons form a circuit with neurons of the VNTB where the MOC somata are.

Reworded: "One week later, retrograde and anterograde staining was observed in the SCC (Fig 2), demonstrating, along with the physiological data, that SCC neurons project to and receive input from MOC neurons, confirming previous findings (Benson et al., 1996; Benson & Brown, 1990; Ye et al., 2000)."

- Line 257 - capitalize "Small" at beginning of line

Corrected

- Figure 4 - why is there no effect of atropine on PL cells in panel A, when there is a clear effect of VNTB stimulation in panel B? Please address.

While these data imply activation of non-cholinergic inputs to primary-like neurons, one alternative explanation is antidromic activation of bushy cells resulting in habituation. This has been commented on at line 458.

- Line 347 - please elaborate on the methods for the application of spike trains
- how applied and where?

Expanded the sentences to clarify the method.

- Line 373 - edit wording of the panel B legend, word incorrect or missing

Corrected. The figure legend from an old version of this figure was still being used.

- Figure 6 data - In panel A and D, the SCs show a number of responses classified as 80 dB, but with an actual dB anywhere from 0 to 80. Please address this, it seems to actually suggest poor intensity coding or the SCs in many instances. What are the criteria for "correct" encoding of intensity?

Your interpretation is correct. There are instances SC spiking does not distinguish intensity as all were classified at 80dB. These would be units with very sharp rate-level function slopes. However, in comparison to choppers and primary-like units, whose spiking does not get classified as above 80 dB *at all* (especially in panel D), these units seem to contribute to SC's overall better population intensity coding. We've added description of this observation to the results.

- Figure 7 results text beginning line 382 - is there an example calculation that was missing? The text refers to an example at 32 kHz, 12% depth, but this is just one point in the fourth panel. Also, the letters are missing from the figure.

Corrected.

- Missing citation of Romero and Trussell 2021, eLife, line 467 - descending input from IC to MOC

Reference added: "Descending modulation of MOC activity would also affect MOC collateral input to CN small cells (Romero & Trussell, 2021), which accurately encode intensity, providing an additional action pathway for auditory attention."

Dear Dr Shore,

Re: JP-RP-2021-282262R1 "Olivocochlear projections contribute to superior intensity coding in cochlear nucleus small cells" by Adam Hockley, Calvin Wu, and Susan Shore

Thank you for submitting your manuscript to The Journal of Physiology. It has been assessed by a Reviewing Editor and by 2 expert referees and I am pleased to tell you that it is considered to be acceptable for publication following satisfactory revision.

Please advise your co-authors of this decision as soon as possible.

The reports are copied at the end of this email. Please address all of the points and incorporate all requested revisions, or explain in your Response to Referees why a change has not been made.

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I hope you will find the comments helpful and have no difficulty returning revisions within 4 weeks.

If you need to check to make sure that your Methods section conforms to the principles of UK regulations, you may wish to refer to Grundy (2015):

Grundy (2015) J. Physiol. 2015 Jun 15;593(12):2547-9 <https://doi.org/10.1113/JP270818>

Your revised manuscript should be submitted online using the links in Author Tasks Link Not Available. This link is to the Corresponding Author's own account, if this will cause any problems when submitting the revised version please contact us.

The image files from the previous version are retained on the system. Please ensure you replace or remove any files that have been revised.

REVISION CHECKLIST:

- Summary data must be reported as mean {plus minus} SD or 95% confidence interval
- All table and figure legends with summary data must include the statistical test used in the table/figure and sample size
- Figures with summary data bars must include individual data points, or box whisker plots when $n > 30$.
- Article file, including any tables and figure legends, must be in an editable format (eg Word)
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To create your 'Response to Referees' copy all the reports, including any comments from the Senior and Reviewing Editors, into a Word, or similar, file and respond to each point in colour or CAPITALS and upload this when you submit your revision.

I look forward to receiving your revised submission.

If you have any queries please reply to this email and the Peer Review Coordinator will be pleased to advise.

If revision is not possible, or if you cannot respond to the requests for change, contact us by return email as soon as

possible, giving reasons for the difficulties. Withdrawal of the manuscript may be necessary in these circumstances, and instruction will be given on how to proceed. Please note that a paper must be withdrawn before it can be submitted to another journal. If any issues remain unresolved please contact the Publications Office at jphysiol@physoc.org

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Yours sincerely,

Ian D. Forsythe
Deputy Editor-in-Chief
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REQUIRED ITEMS:

Please could authors add the following:

- 1) Euthanasia details (type/amount used etc)
- 2) Data Availability statement

Thank you,

Acting Peer Review Co-ordinator.

EDITOR COMMENTS

Reviewing Editor:

Thank you for including the final requested changes.

Senior Editor:

Comments for Authors to ensure the paper complies with the Statistics Policy:
Please could you read our instructions to authors for data and figure presentation.

Figures and/or legends needs to specify N, use exact P values (not *) and

If the statistical summary document has errors please describe what is incorrect:

The SSD is incomplete and confusing.

As an indication of how to do this - the SSD should be capable of being read without detailed reference to the text. It is also intended to be a Key Points for the numerical aspects of the manuscript.

Specify animal age. Male and/or female?

The hypothesis as stated is trivial - please amend to better capture your interesting question and intent.

The conclusion from each question is missing

Define abbreviations in the comments section, explain what you's done. Make the variables clear (e.g. could include ranges of animal age/sex)

Please take advice on t-test you cannot just test multiple times - use your question to define what you are testing and then make the result clear. There is no point in showing that umpteen intensity values are different, if the question is about pharmacology... as in Qs 2-4.

Why are all the numbers in entered commas?

From question 11 onwards perhaps it would be clearer to have PL, CH and Sc on different lines, so it is obvious which p values applies.....

Comments to the Author:

Congratulations on an interesting manuscript. Please take some advice on how to present the data and statistics well; it is not appropriate to do multiple t-tests on an intensity plot, when you are testing the action of a drug. Your data presentation is a major outstanding issue, which will require some updating of figure legends, figures and SSD. You need to phrase your statistical tests in terms of the question and the conclusion. Please take some advice on how to present this well; it is not appropriate to do multiple t-tests on an intensity plot, when you are testing the action of a drug, for example.

REFeree COMMENTS

Referee #1:

The authors have been diligent in addressing my comments. I congratulate them on an improved manuscript.

Referee #2:

The authors have addressed my concerns. The addition of examples from individual cells is particularly nice.

END OF COMMENTS

1st Confidential Review

09-Oct-2021

Revision checklist:

If you need to check to make sure that your Methods section conforms to the principles of UK regulations, you may wish to refer to Grundy (2015):

Grundy (2015) J. Physiol. 2015 Jun 15;593(12):2547-9 <https://doi.org/10.1113/JP270818>

The manuscript now complies with the checklist presented by Grundy 2015.

Acting Peer Review Co-ordinator:

1) Euthanasia details (type/amount used etc)

This is included on line 125

2) Data Availability statement

This has been added on line 35

Senior Editor:

Please could you read our instructions to authors for data and figure presentation.

This has been read and the manuscript complies with the instructions.

Figures and/or legends needs to specify N, use exact P values (not *) and

N's have been added to figure legends.

The SSD is incomplete and confusing.

More detail has been added to the SSD.

As an indication of how to do this - the SSD should be capable of being read without detailed reference to the text. It is also intended to be a Key Points for the numerical aspects of the manuscript.

Specify animal age. Male and/or female?

Precise animal ages are unknown due to our external supplier. All animals (11M, 14F) weighed 280-800g at the time of experiments, as per line 107. N's in the SSD refer to number of units.

The hypothesis as stated is trivial - please amend to better capture your interesting question and intent.

Underlying hypothesis has been updated to: Medial olivocochlear collateral projections to small cells of the cochlear nucleus counteract the effect of cochlear suppression, allowing small cell firing to accurately represent stimulus intensity.

The conclusion from each question is missing

Conclusions have been added to each question.

Define abbreviations in the comments section, explain what you's done. Make the variables clear (e.g. could include ranges of animal age/sex)

Abbreviations have been added.

Please take advice on t-test you cannot just test multiple times - use your question to define what you are testing and then make the result clear. There is no point in showing that umpteen intensity values are different, if the question is about pharmacology... as in Qs 2-4.

The multiple t-tests on intensity plots with drug in fig 4 have been removed. Instead, we have now statistically compared peak firing rate, and RLF slope between 50 and 90 dB SPL. The figure, legend, results text and SSD have been updated to reflect this.

Why are all the numbers in entered commas?

This has been resolved.

From question 11 onwards perhaps it would be clearer to have PL, CH and Sc on different lines, so it is obvious which p values applies.....

The text for p values for Q11-14 has been updated (e.g. SCvsPL). This clarifies which p values are for which comparisons.

Comments to the Author:

Congratulations on an interesting manuscript. Please take some advice on how to present the data and statistics well; it is not appropriate to do multiple t-tests on an intensity plot, when you are testing the action of a drug. Your data presentation is a major outstanding issue, which will require some updating of figure legends, figures and SSD. You need to phrase your statistical tests in terms of the question and the conclusion. Please take some advice on how to present this well; it is not appropriate to do multiple t-tests on an intensity plot, when you are testing the action of a drug, for example.

Thank you for this insight, we have removed multiple t-test from the intensity plots when testing the action of drug or electrical stimulation.

Dear Dr Shore,

Re: JP-RP-2021-282262R2 "Olivocochlear projections contribute to superior intensity coding in cochlear nucleus small cells" by Adam Hockley, Calvin Wu, and Susan Shore

I am pleased to tell you that your paper has been accepted for publication in The Journal of Physiology.

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Yours sincerely,

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EDITOR COMMENTS

Senior Editor:

Thank you for these final changes. Please note that figures 4 and 5 have some minor issues - there is a 'point box' inserted in figure 4 and the stars are still present in figure 5; these can be amended in the proofs.

END OF COMMENTS

2nd Confidential Review

02-Nov-2021